

KHAOULA BELHAJ

ENGINEERING PORTFOLIO

*From networks and cloud infrastructure to cybersecurity,
software and intelligent systems.*

CLOUD · NETWORKS · CYBERSECURITY · SOFTWARE ·
TELECOMMUNICATIONS · AI

ENSA FEZ

ENGINEERING DEGREE IN COMMUNICATING SYSTEMS & COMPUTER
SECURITY

2024 - 2027

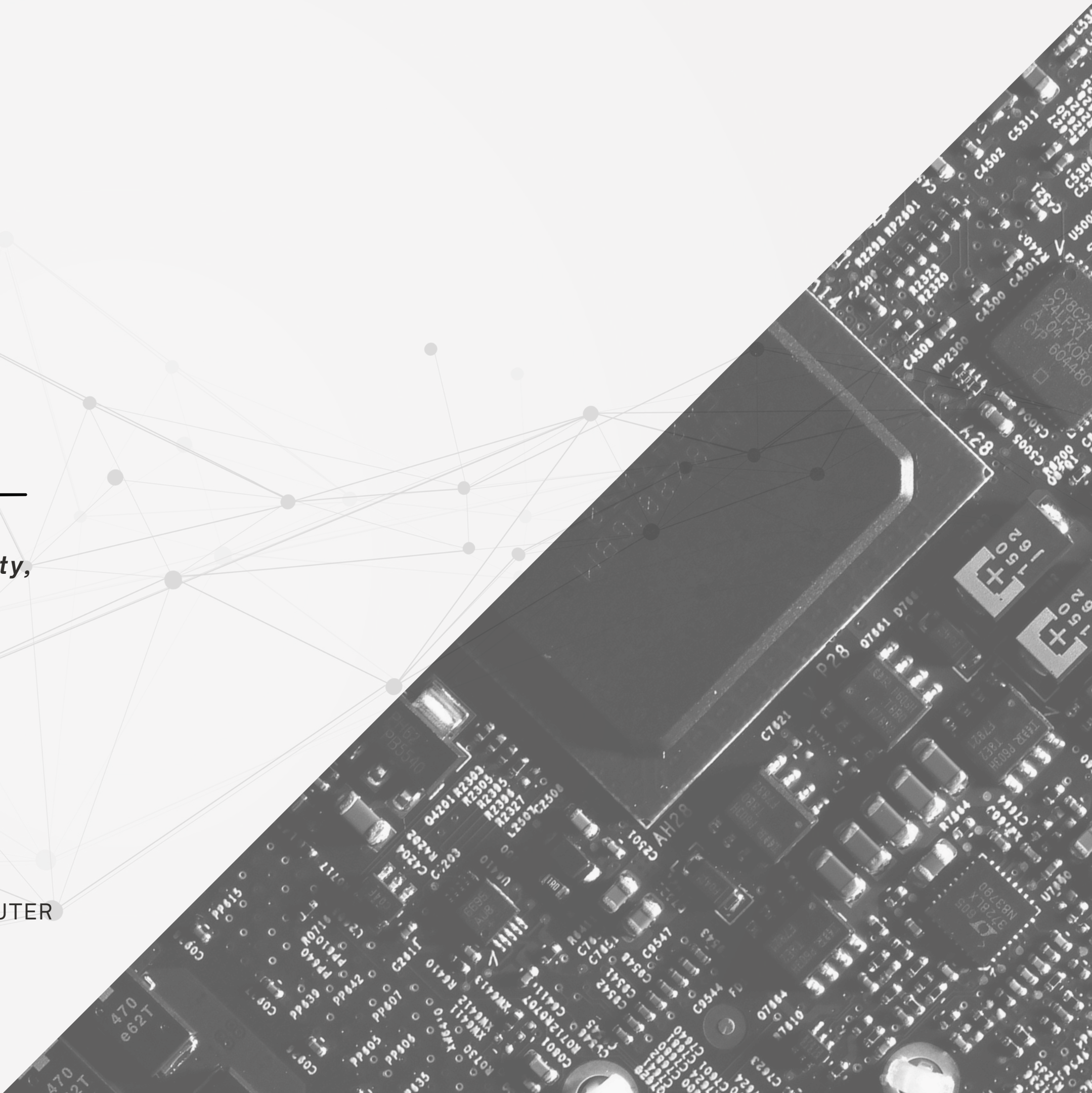


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MY ENGINEERING JOURNEY

FROM COMMUNICATION TO INTELLIGENT AND SECURE SYSTEMS

My engineering journey has been shaped by a multidisciplinary approach to digital systems. Starting with networks and telecommunications, I developed a strong understanding of how systems communicate and operate. I progressively expanded this foundation into software development, cloud infrastructure and cybersecurity, while exploring artificial intelligence, automation and embedded systems.

This progression has allowed me to build a broader engineering perspective — from designing and connecting systems to deploying, securing and improving them.

01

CONNECT

Understanding how systems communicate

Networks · Telecommunications · Routing & Switching

02

BUILD

Turning ideas into digital solutions

Software · Web · Mobile · Databases

03

SCALE

Designing scalable infrastructures

Cloud · Virtualization · Distributed Systems

04

PROTECT

Securing systems and data

Cybersecurity · Cryptography · Intrusion Detection

05

AUTOMATE

Making systems smarter and more efficient

Python · Artificial Intelligence · Machine Learning

06

ENGINEER

Bringing technologies together to solve complex problems

System Architecture · Integration · Problem Solving

I see modern engineering as the connection between communication, software, infrastructure, intelligence and security.

I am **Khaoula BELHAJ**, a final-year Engineering student at ENSA Fez, specializing in Communicating Systems and Computer Security.

What defines my approach is not only the technologies I work with, but the way I approach problems. I enjoy understanding the requirements behind a technical challenge, breaking complex problems into structured solutions, and turning ideas into functional and reliable results.

Through my academic projects and professional experiences, I have worked in different technical environments, from network and systems infrastructure to web and mobile application development. These experiences have strengthened my ability to adapt to new environments, learn independently and collaborate effectively while remaining focused on the objectives of a project.

I am naturally curious, analytical and solution-oriented. I enjoy exploring new technologies, understanding how systems work and continuously developing my technical skills through hands-on projects and real-world challenges.

ABOUT ME

CAREER GOAL

I am seeking a PFE / Final-Year Internship where I can contribute to a real technical project, take on meaningful responsibilities and further develop my engineering expertise through hands-on experience.

I am particularly interested in opportunities related to Cloud, Network Engineering, Cybersecurity and IT Infrastructure, while remaining open to multidisciplinary engineering projects. My goal is to contribute, learn and grow within a professional environment, with the potential to continue the collaboration beyond the internship when there is a strong mutual fit.

CURRENT FOCUS

-  CLOUD & INFRASTRUCTURE
-  NETWORK ENGINEERING
-  CYBERSECURITY
-  SOFTWARE & MOBILE
-  AI & AUTOMATION



CHAPTER 01

CONNECT

Networks & Telecommunications

NETWORK ARCHITECTURE

ROUTING & SWITCHING

NETWORK SERVICES

MOBILE COMMUNICATIONS

DIGITAL COMMUNICATIONS

IoT



MULTI-SITE FINANCIAL NETWORK

Enterprise Network Architecture & GNS3 Simulation

Project NEXUS is a high-availability banking network architecture designed and simulated in GNS3. The infrastructure connects a primary Data Center, a Disaster Recovery site, an administrative headquarters and a remote banking agency.

The project focuses on service continuity, network resilience, security and high availability through structured routing, redundancy and network segmentation.

OBJECTIVES

- **High Availability**

Eliminate single points of failure and maintain critical services.

- **Network Resilience**

Provide continuity and rapid recovery in case of an incident.

- **Secure Segmentation**

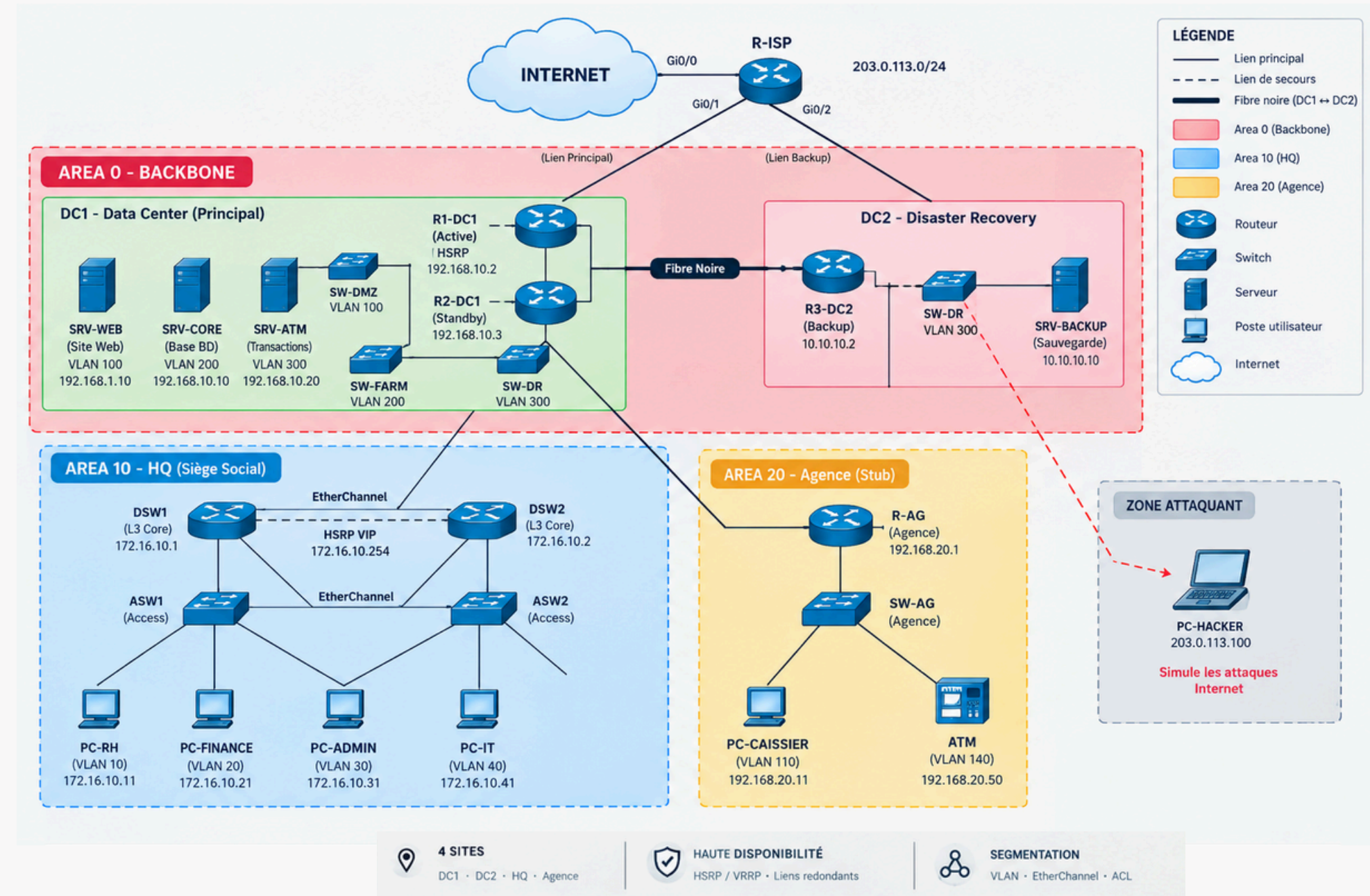
Separate functional domains using VLANs and access control.

- **Inter-Site Connectivity**

Ensure reliable communication between DC1, DC2, HQ and the remote agency

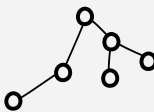
TECHNOLOGIES

GNS3 • OSPF Multi-Area • HSRP / VRRP • VLAN • EtherChannel / LACP • ACL • Inter-VLAN Routing



Hierarchical Design & High Availability

ARCHITECTURE



MULTI-AREA OSPF

The network is structured into three OSPF areas: Area 0 as the backbone, Area 10 for the headquarters and Area 20 for the remote banking agency. This hierarchical organization provides structured inter-site routing and limits routing complexity.



HIGH AVAILABILITY

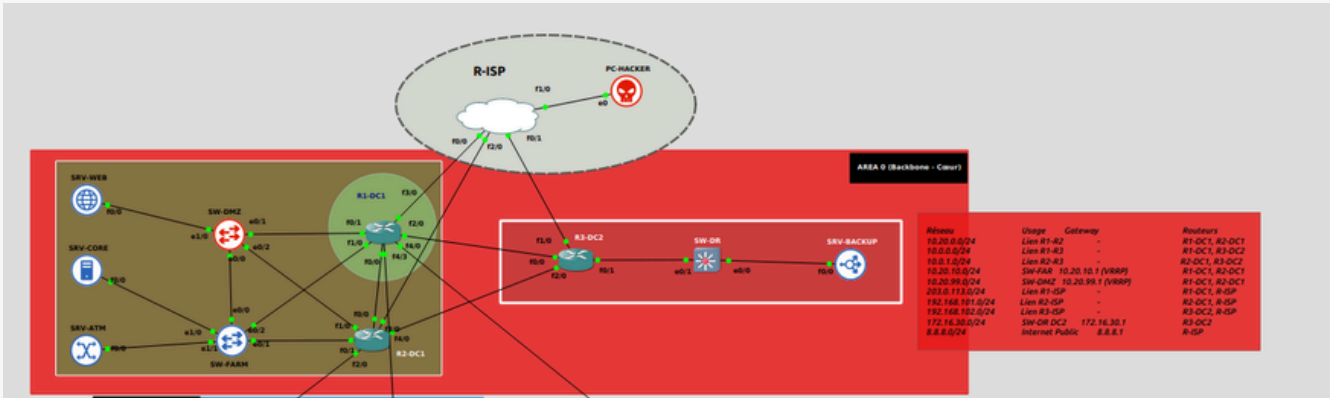
REDUNDANT CORE



DISASTER RECOVERY

DC1 → Fibre Noire → DC2

A dedicated connection links the primary Data Center to the Disaster Recovery site, supporting continuity in case of a major incident.



Data Center & Core (AREA 0)

VLAN	Name	Status	Ports
1	default	active	Et0/3, Et1/3, Et2/0, Et2/1, Et2/2, Et2/3, Et3/0, Et3/1, Et3/2, Et3/3
10	DIRECTION	active	
20	RH	active	
30	FINANCE	active	Et1/0
40	AUDIT	active	
50	IT-ADMIN	active	Et1/1
60	EMPLOYES	active	
70	VOIP	active	
80	ATM-BACKOFFICE	active	Et1/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

ASW2#sh run
Building configuration...

VLAN

from LOADING to FULL, Loading Done

R1-DC1#sh vrrp brief

Interface	Grp	Pri	Time	Own	Pre	State	Master addr	Group addr
Fa1/0	10	150	3414	Y	Master		10.20.10.2	10.20.10.1
Fa0/1	99	150	3414	Y	Master		10.20.99.2	10.20.99.1

R1-DC1#

Gateway Redundancy (HSRP / VRRP)

Number of channel-groups in use: 1
Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Et0/1(P) Et0/2(P)

EtherChannel / LACP

 OSPF Multi-Area Routing	 HSRP / VRRP Gateway Redundancy	 VLAN Network Segmentation	 ETHERCHANNEL Link Aggregation
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Testing the Network Architecture

The simulated infrastructure was validated through connectivity, routing, redundancy and switching tests across the different network areas.

01

ROUTING

- OSPF Neighbors
- Routing Table
- OSPF Database

```
R-AG#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:37	172.16.20.1	FastEthernet0/0

OSPF Neighbors

```
R-AG#show ip ospf database
```

OSPF Router with ID (20.20.20.1) (Process ID 1)

Router Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	845	0x80000002	0x00201B	2
20.20.20.1	20.20.20.1	849	0x80000002	0x00C77C	5

Summary Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
0.0.0.0	1.1.1.1	851	0x80000001	0x0093A6

OSPF Database

02

REDUNDANCY

- HSRP / VRRP
- Master / Backup States
- Gateway Availability

```
DSW2#sh standby brief
```

P indicates configured to preempt.

Interface	Grp	Pri	P State	Active	Standby	Virtual IP
Vl10	1	100	P Standby	10.10.10.2	local	10.10.10.1
Vl20	2	100	P Standby	10.10.20.2	local	10.10.20.1
Vl30	3	100	P Standby	10.10.30.2	local	10.10.30.1
Vl40	4	100	P Standby	10.10.40.2	local	10.10.40.1
Vl50	5	100	P Standby	10.10.50.2	local	10.10.50.1
Vl60	6	100	P Standby	10.10.60.2	local	10.10.60.1
Vl70	7	100	P Standby	10.10.70.2	local	10.10.70.1
Vl80	8	100	P Standby	10.10.80.2	local	10.10.80.1

DSW2#

Gateway Redundancy

03

SWITCHING

- VLANs
- Trunking
- EtherChannel / LACP

```
DSW2#sh etherchannel summary
```

Flags: D - down P - bundled in port-channel
I - stand-alone S - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use N - not in use, no aggregation
f - failed to allocate aggregator

M - not in use, minimum links not met
m - not in use, port not aggregated due to minimum links not met
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators: 1

Group	Port-channel	Protocol	Ports
2	Po2(SU)	LACP	Et0/1(P) Et0/2(P)

Link Aggregation

04

CONNECTIVITY

- ICMP Tests
- Inter-site Connectivity
- ATM Connectivity

```
ATM-Machine> ping 172.16.22.21
```

172.16.22.21 icmp_seq=1 ttl=64 time=0.001 ms
172.16.22.21 icmp_seq=2 ttl=64 time=0.001 ms
172.16.22.21 icmp_seq=3 ttl=64 time=0.001 ms
172.16.22.21 icmp_seq=4 ttl=64 time=0.001 ms
172.16.22.21 icmp_seq=5 ttl=64 time=0.001 ms

```
ATM-Machine> ping 172.16.21.11
```

172.16.21.11 icmp_seq=1 timeout
84 bytes from 172.16.21.11 icmp_seq=2 ttl=63 time=21.342 ms
84 bytes from 172.16.21.11 icmp_seq=3 ttl=63 time=16.848 ms
84 bytes from 172.16.21.11 icmp_seq=4 ttl=63 time=23.322 ms
84 bytes from 172.16.21.11 icmp_seq=5 ttl=63 time=23.067 ms

```
ATM-Machine> ping 10.10.10.10
```

10.10.10.10 icmp_seq=1 timeout
10.10.10.10 icmp_seq=2 timeout
*203.0.113.2 icmp_seq=3 ttl=254 time=400.293 ms (ICMP type:11, code:0, TTL expired in transit)
*203.0.113.2 icmp_seq=4 ttl=254 time=650.561 ms (ICMP type:11, code:0, TTL expired in transit)
*203.0.113.2 icmp_seq=5 ttl=254 time=521.309 ms (ICMP type:11, code:0, TTL expired in transit)

END-TO-END Connectivity



PROJECT RESULT

The architecture validated the intended engineering choices and provided a solid basis for high availability, segmentation and resilience.

CHAPTER 02

BUILD

Software & Mobile Development

WEB DEVELOPMENT

JAVA / JEE

MOBILE APPLICATIONS

REST APIs

DATABASES

UI/UX

SOFTWARE IMPLEMENTATION

02.01 VEHICLE RENTAL PLATFORM

Web & Mobile Vehicle Rental Management Platform

PROJECT OVERVIEW

A digital platform developed for El Bakkali Prestige Auto, a vehicle-rental agency in Meknes. The solution combines a customer-facing web platform for vehicle browsing and online reservations with a mobile application dedicated to internal management. Both applications rely on a PHP REST API and a centralized MySQL database.

THE CHALLENGE

The project addressed a rental-management process that relied largely on manual handling of reservations, vehicles, customers and invoicing. The objective was to centralize these operations through a connected digital platform.

TECHNOLOGY STACK

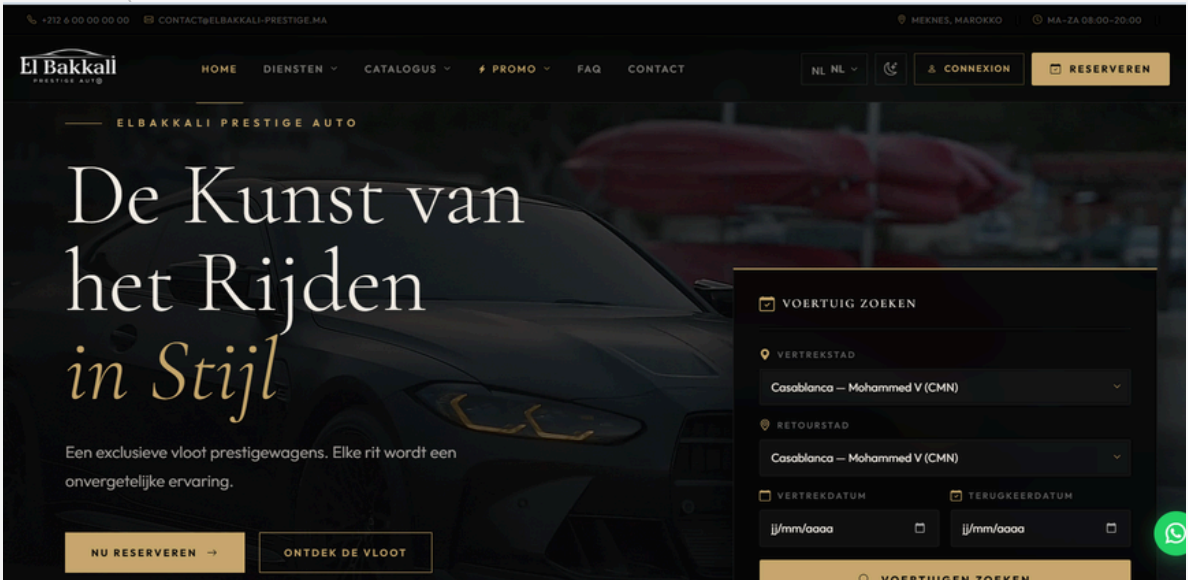
 WEB	 MOBILE	 BACKEND	 DATABASE	 TOOLS
PHP HTML5/CSS3 JavaScript Bootstrap 5	Flutter Dart	PHP REST API	MySQL SQL	VS Code XAMPP Android Studio Git

SOLUTION

WEB PLATFORM

Customer-facing experience

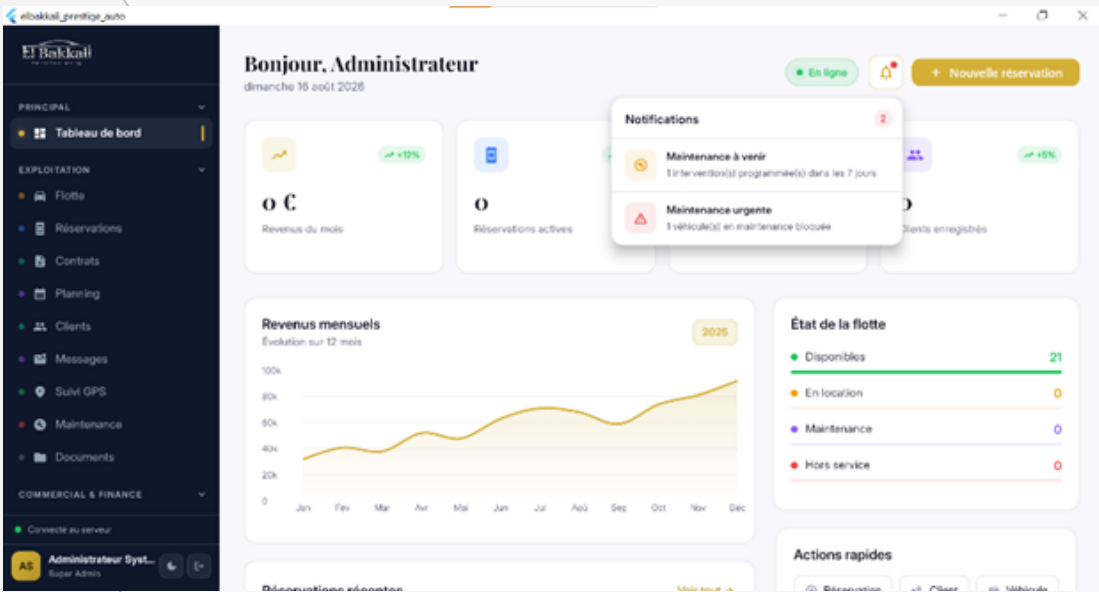
- Vehicle catalogue
- Vehicle filtering
- Online reservation
- Options selection
- Customer information
- Payment
- Confirmation & PDF receipt



MOBILE APPLICATION

Administrative management

- Vehicles
- Customers
- Reservations
- Payments
- Maintenance
- Contracts
- Archives



WEB PLATFORM

02.01

Customer-Facing Rental Experience

The web platform provides customers with a complete online vehicle rental experience, from browsing available vehicles to completing a reservation. It brings together vehicle discovery, availability, rental options, customer information, payment and reservation confirmation within a structured and responsive interface.

RESERVATION FLOW

- 01

DATES & LOCATIONS
- 02

VEHICLE
- 03

OPTIONS
- 04

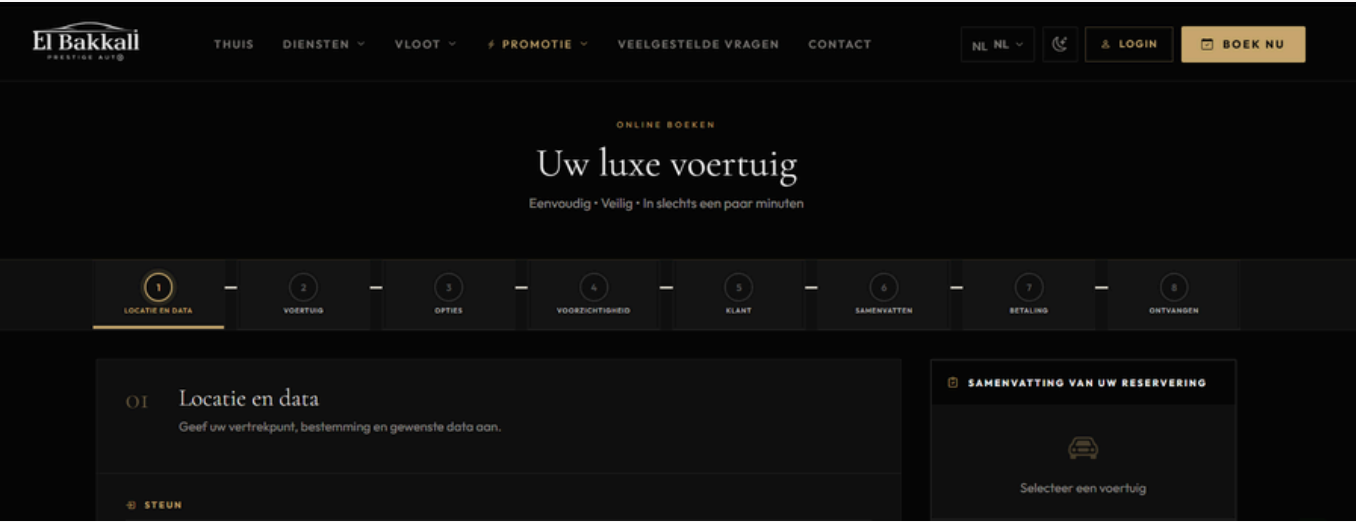
DEPOSIT
- 05

CUSTOMER INFORMATION
- 06

SUMMARY
- 07

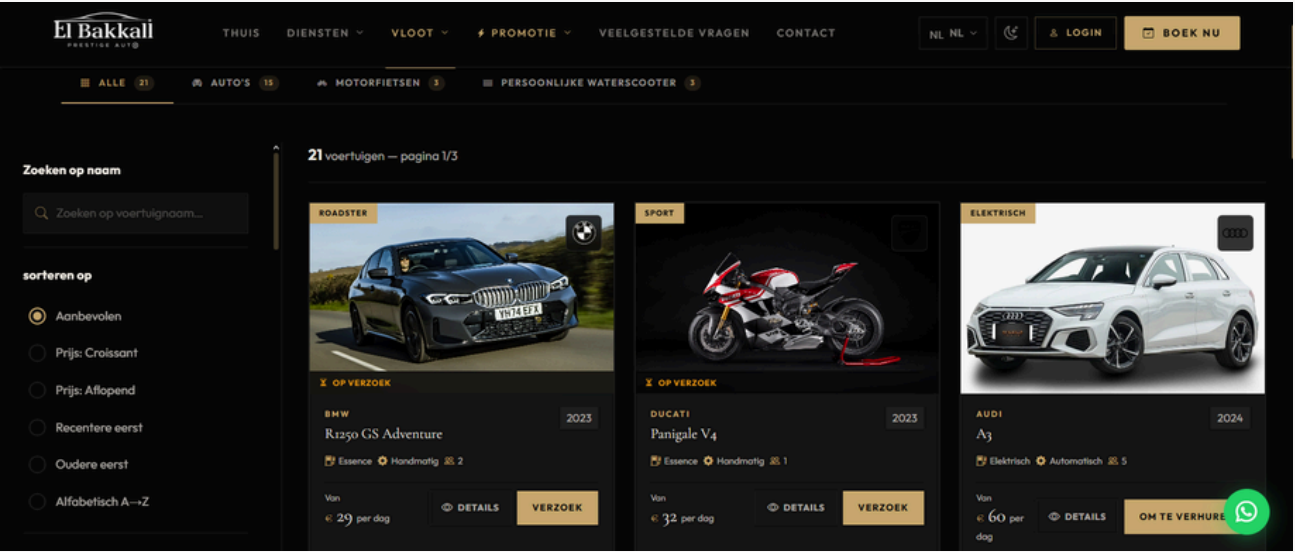
PAYMENT
- 08

CONFIRMATION & RECEIPT



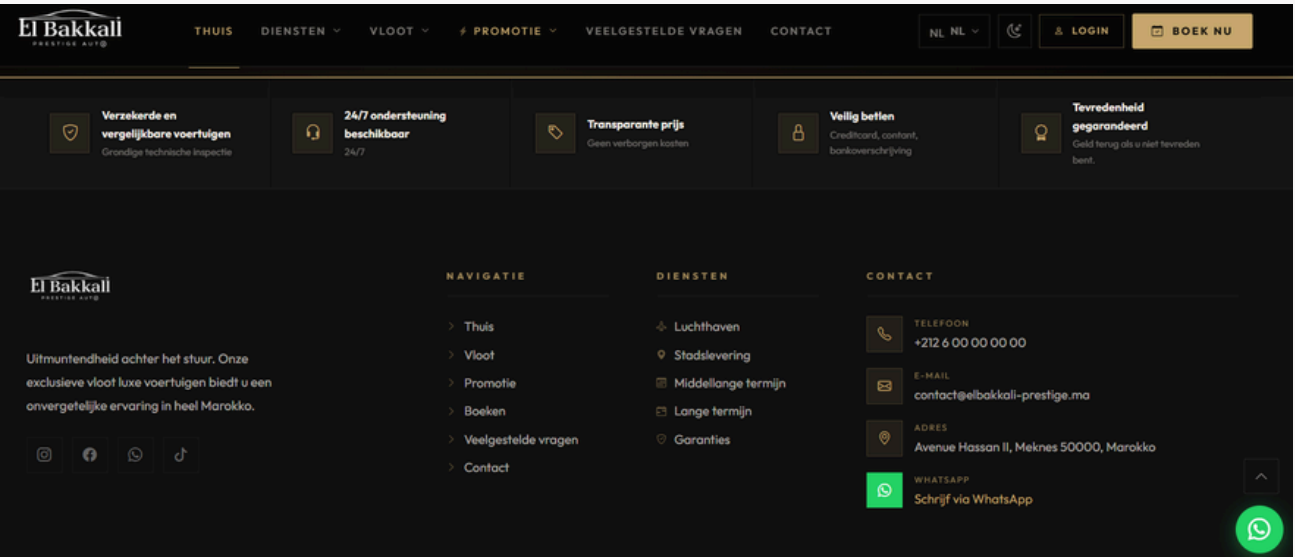
CATALOGUE

The vehicle catalogue dynamically presents vehicle information including photography, brand, model, type, seating capacity, transmission, year, daily rate and availability. A filtering system allows vehicles to be searched according to criteria such as type, brand, price and availability.



RESPONSIVE DESIGN

The web interface was designed to adapt to different screen formats, supporting a responsive browsing experience.

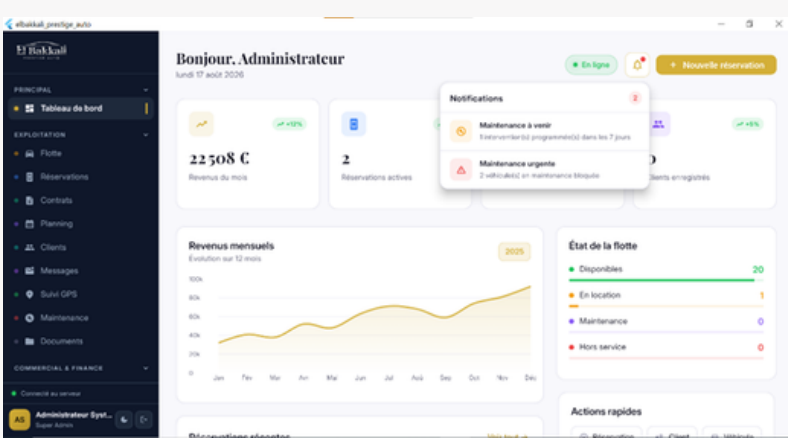


Administrative Management on the Go

The mobile application was developed with Flutter and Dart to centralize the internal management of the rental agency. Designed for administrators and employees, it provides access to vehicles, customers, reservations, payments, maintenance, contracts, documents and archives from a smartphone.

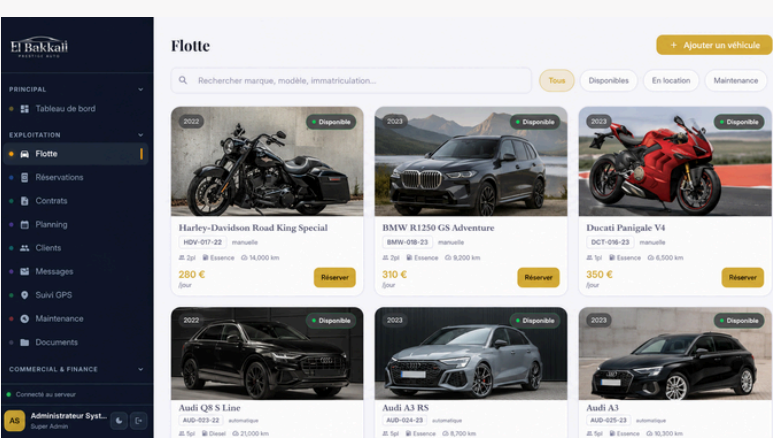
SECTION 01 — DASHBOARD

The dashboard provides a centralized view of the agency's daily activity, including key statistics, recent reservations and vehicle information. Data displayed in the application is retrieved through the backend REST API.



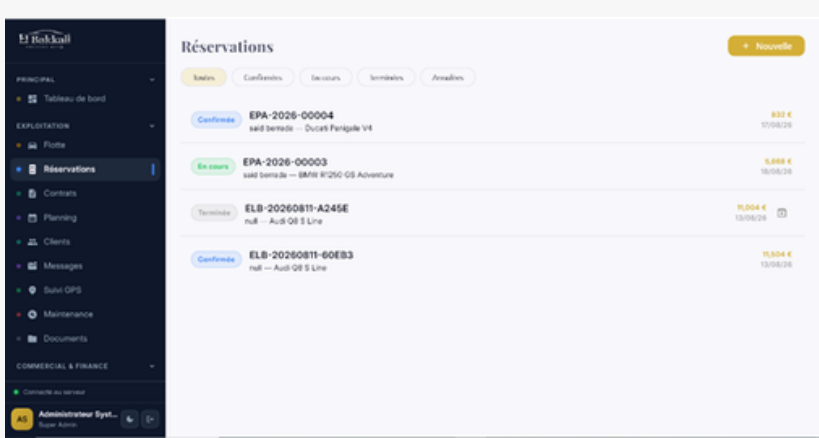
SECTION 02 — FLEET & CUSTOMER MANAGEMENT

The application provides dedicated modules for managing the vehicle fleet and customer information. Vehicle records include key operational data such as model, registration, mileage, pricing and status, while customer records are linked to reservations and rental documents.



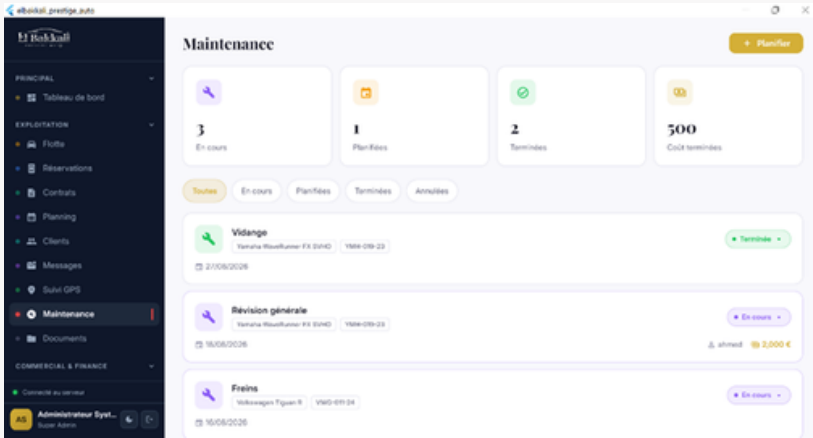
SECTION 03 — RESERVATIONS & OPERATIONS

Reservations can be consulted and managed directly from the application. Each reservation brings together the customer, vehicle, rental period, options and payment information, providing the staff with a centralized view of ongoing rental operations.



SECTION 04 — MAINTENANCE & ARCHIVES

Vehicle maintenance operations can be planned and monitored through dedicated statuses, including scheduled, ongoing, completed and cancelled interventions.

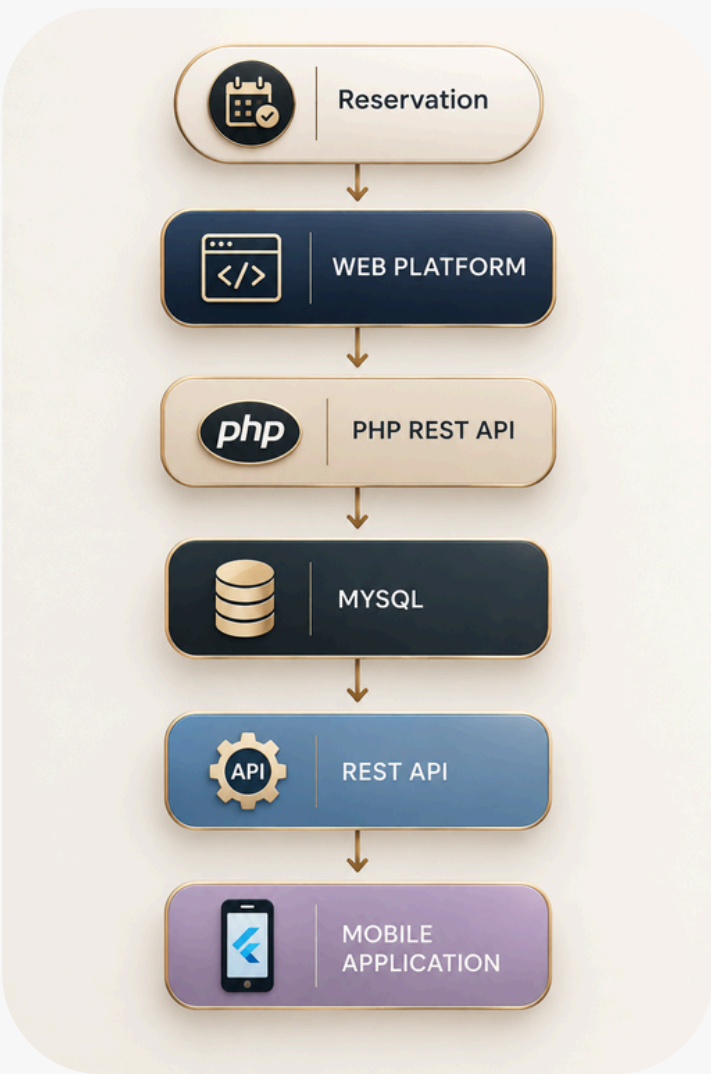


From Web to Mobile

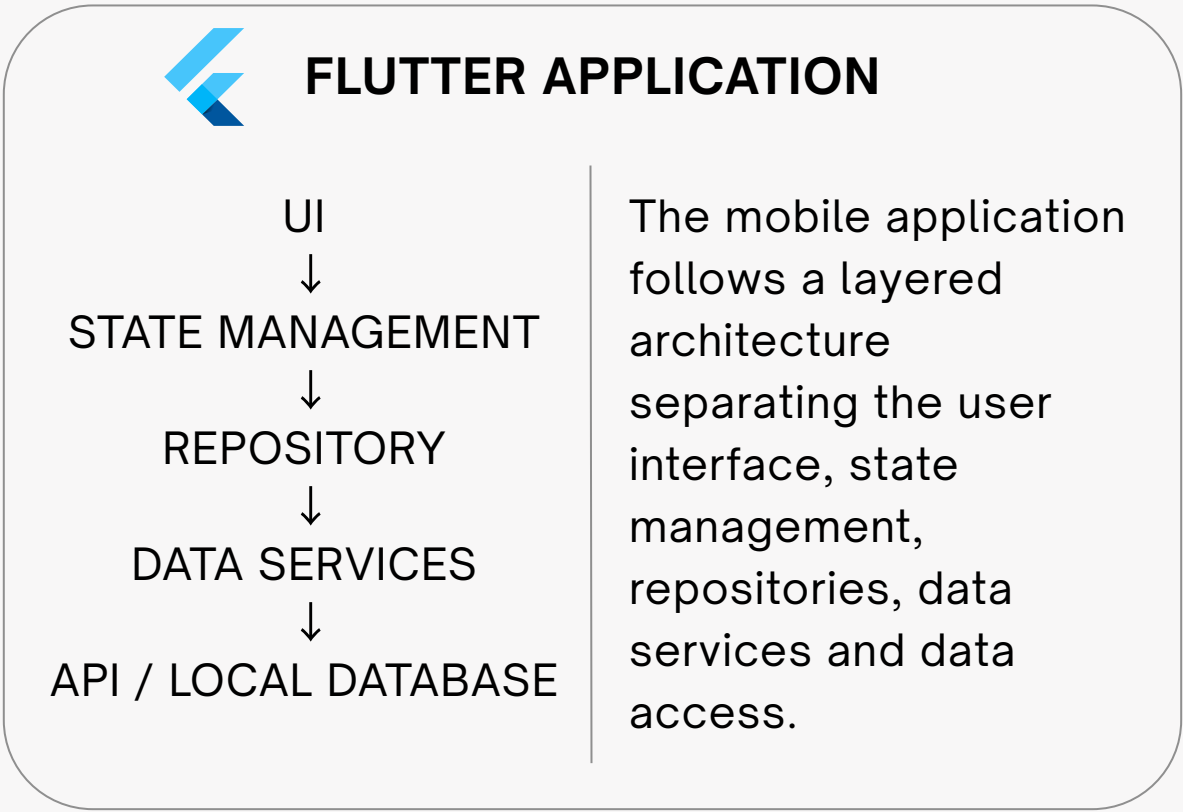
The platform connects the customer-facing web application and the administrative mobile application through a PHP REST API and a shared MySQL database. Both interfaces remain independent while exchanging data through the backend.

DATA FLOW

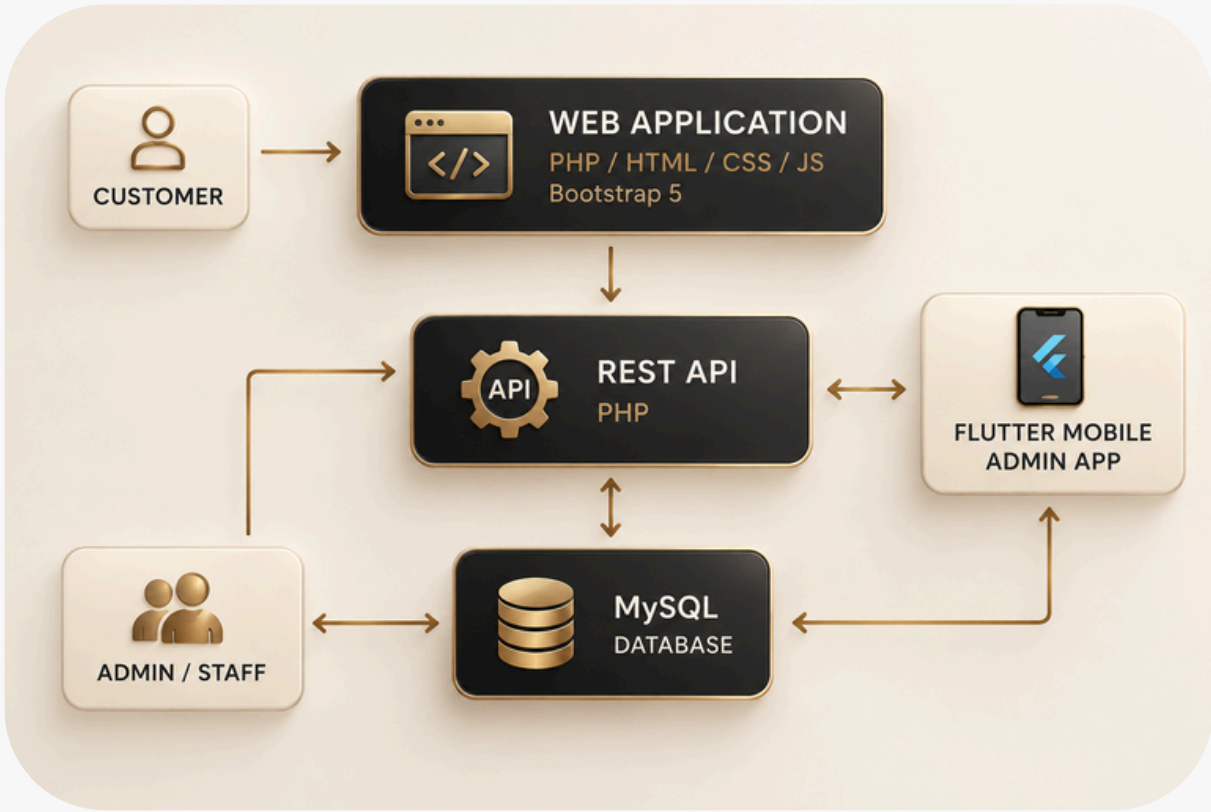
WEB → BACKEND → MOBILE



MOBILE ARCHITECTURE



SYSTEM ARCHITECTURE



TECHNOLOGY STACK

FLUTTER	DART	REST API	JSON	JWT	PHP	Local Storage	MySQL
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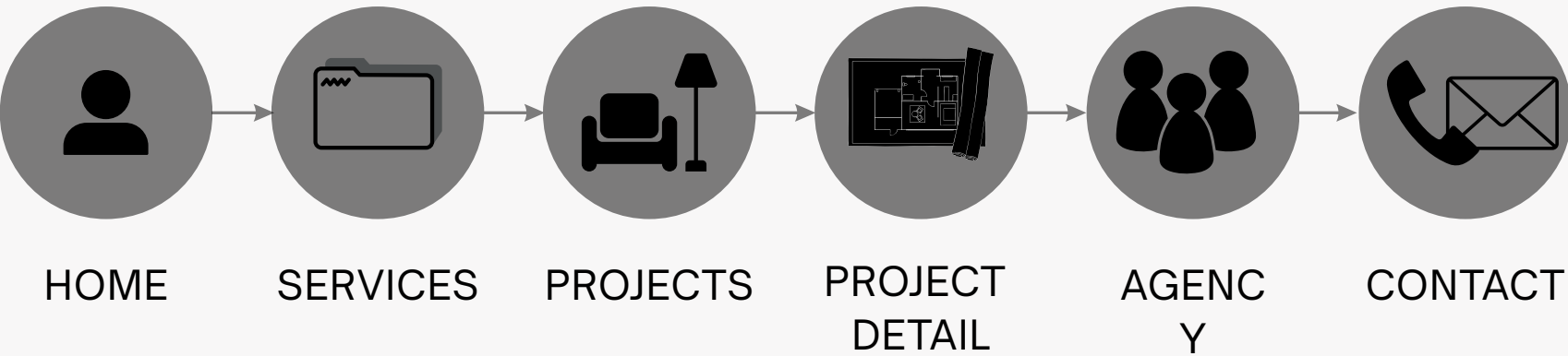
02.02 ARCHITECT WEBSITE

Digital Portfolio & Web Experience for an Architecture Studio

Designed and developed a responsive web platform for an architecture and interior design studio, transforming its visual identity and project portfolio into a structured digital experience.

The website brings together the studio presentation, services, architectural projects, individual project pages and contact experience within a cohesive interface designed around visual storytelling and intuitive navigation.

WEBSITE STRUCTURE



Discover → Explore → Understand → Connect

PROJECT CONTRIBUTION

Designed and developed the website interface and front-end experience, including its page structure, responsive layouts, interactive components and project presentation system.



PROJECT SCOPE

- **Web Design**
Interface structure, visual hierarchy and presentation.
- **Front-End Development**
Implementation using HTML, CSS and JavaScript.
- **Interactive Experience**
Dynamic project presentation, navigation and image galleries.
- **Responsive Interface**
Adaptation of the experience to different screen sizes.

BUILDING THE EXPERIENCE

02.02

Implementation

A responsive front-end experience built with HTML, CSS and JavaScript, combining dynamic project presentation, interactive galleries and a functional contact form.

TECHNOLOGY STACK



HTML

Structure & Content

Semantic page structure and organization of the website's content.



CSS

Visual System & Responsive Layout

Custom layouts, typography, visual styling, animations and responsive behaviour.



JAVASCRIPT

Interaction & Dynamic Content

Navigation, project rendering, gallery interactions, mobile behaviour and user interface logic.



EMAILJS

Contact Integration

Client-side integration used to submit contact form information.

CORE FUNCTIONALITY

01 — DYNAMIC PROJECTS

Project content displayed dynamically from structured JavaScript data.

02 — INTERACTIVE GALLERY

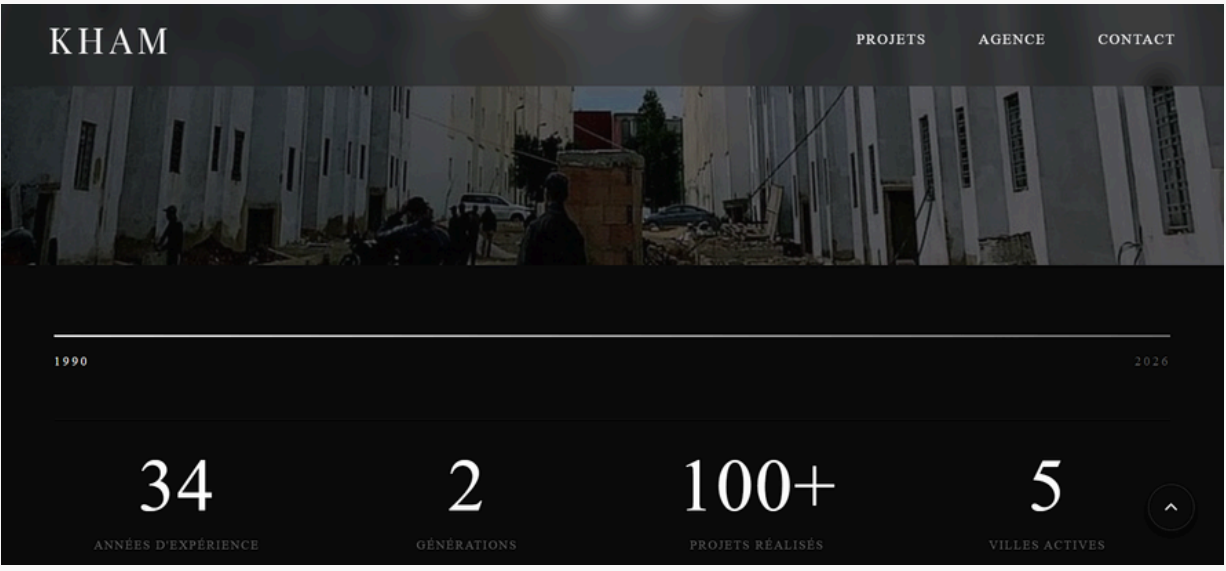
Lightbox with image navigation and counter.

03 — RESPONSIVE EXPERIENCE

Desktop and mobile layouts with dedicated navigation interactions.

04 — CONTACT FORM

Form validation and email submission through EmailJS.



TECHNICAL FLOW

